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Erratum

Erratum to "Mathematical modelling of hepatic lipid metabolism, Mathematical Biosciences 262 (2015) 167–181"

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The publisher regrets truncating Table 3
The publisher would like to apologise for any inconvenience caused.

Table 3
Parameter values used in simulations.

Parameter	Value	Description	Source
α_A	15.6 l	Volume of adipose tissue	A 70 kg individual with a BMI of 21.5 kg/m^2 (healthy) has an estimated body fat of 20% (1994 National Health and Nutrition Examination Study). Conversion to volume is obtained from an adipose tissue density of 0.9 kg/l.
α_B	5 l	Volume of blood	Ref. [19].
α_L	1.6 l	Volume of the liver	Ref. [5].
α_M	26.4 l	Volume of skeletal muscle	A 70 kg individual has an estimated 40% body weight in skeletal muscle. Conversion to volume is obtained from a skeletal muscle tissue density of 1.06 kg/l.
β_6	31.6	Rate of liver DNL from pyruvate	Refs. [31], [32].
β_f	0.117 mmol/min	Adipose release of TAG to NEFA	Approximated from data in Bickerton [8,9].
β_l	12 l/min	Liver glycogenolysis	Approximations from considering data in Taylor [55], solving glycogenolysis and glycogenesis rate equations simultaneously.
β_m	82.5 l/min	Muscle glycogenolysis	Ref. [31].
μ_{amp}	1.8	Notional AMP/ADP creation rate	A typical energy expenditure of 10 MJ/day is assumed [19]. 15% of this is satisfied by protein consumption [19], and 2025 from brain usage, leaving 6475 kJ. Conversion for molar density and energy density gives 1.8.
μ_b	0.133 mmol/min	Lactate production by red blood cells	Ref. [19].
μ_e	0.420 mmol/min	Muscle TAG breakdown to FFA	Small, set to the fasting FFA oxidation rate.
μ_s	7.19×10^6 l mmol^{-1} min^{-1}	Muscle FFA esterification to TAG	Rate chosen such that the skeletal muscle FFA concentration is 0.57 mmol/l.
μ_1	0.588 mmol/min	Plasma glucose usage	Total rate is the sum of brain and red blood cell used. Brain usage of plasma glucose is 0.458 mmol/min (120 g/day). Red blood cell usage is 0.133 (25 mg/min) [19].
μ_3	7.839×10^7 l mmol^{-1} min^{-1}	Muscle G-6-P usage	Approximate glucose usage in periphery is 3.2 MJ [19]. Rate takes into account G-6-P and insulin concentration, and FFA usage. Ratio of μ_3 and μ_4 determines usage.
μ_4	100 l/min	Muscle FFA usage	Approximate FFA usage in periphery is 6.1 MJ [19]. Rate takes into account muscle concentration, and glucose usage. Ratio of μ_3 and μ_4 determines usage.

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Table 3 (*continued*)

Parameter	Value	Description	Source
c_0	0.1 mmol/l	Small parameters	Chosen to be $<O(0.01)$ of glycogen concentration
cc	2.5 mmol/l	Range of glucose concentrations over which excess insulin secretion occurs	Obtained from a simple least squares data fit of experiments in Ref. [22] (Fig. 1A).
d_{BA}	0.3	Adipose uptake of glucose	Rate chosen such that the adipose glucose steady state concentration is 2.54 mmol/l [56].
k_{10}	0.625 mmol/l	Affinity for hydrolysis of TAG to secretory pool	Arbitrary small parameter. Chosen to be $<O(0.01)$ of TAG pool.
k_{11}	48 mmol/min	Basal insulin secretion rate	Obtained from a simple least squares data fit of experiments in Ref. [22] (Fig. 1A).
k_{12}	0.2	Increased fraction of VLDL1 secretion by insulin	Simple least squares fitting of multiple data sets [3] [4] [39] [20].
k_{13}	15 mmol/l	Rate at which insulin modifies the fraction of VLDL1/VLDL2 secretion	See k_{12}.
k_{14}	0.6	Basal VLDL1 secretion fraction	See k_{12}.
k_{22}	48 mmol/min	Excess insulin secretion rate due to glucose stimulation	Obtained from a simple least squares data fit of experiments in Ref. [22] (Fig. 1A).
k_5	$8.23 \times 10^7 \text{mmol}^{-1}$	Flux control coefficient for insulin inhibition of FFA oxidation	Estimated from experiments in Westerbacka [60].
k_6	0.3 mmol/l	Affinity for VLDL2 TAG secretion through secretory pathway	Chosen to be $<O(0.01)$ of FFA concentration
$k_{6\,l}$	4 l/min	Liver glucose dephosphorylation rate	Ref. [31]
k_{6p}	$6.56 \times 10^8\ \text{l}^2\ \text{mmol}^{-1}\ \text{min}^{-1}$	Muscle G-6-P to pyruvate conversion rate	Chosen such that steady state concentration of muscle G-6-P is 0.13 mmol/l.
k_7	0.759 l/min	Max rate of FFA oxidation	For a liver TAG concentration of 45 mmol/l at steady state, the oxidation rate is set to maintain this. Parameter is adjusted for insulin inhibition by fitting to experimental data in Westerbacka et al. [60]. Note, we assume ketone production is proportional to FFA oxidation.
k_8	0.625 mmol/l	Affinity for esterification of FFA to TAG	Small parameter chosen to be $<O(0.1)$ of FFA concentration.
k_9	43.583 mmol/l	Affinity of additional bulk lipidation	See v_9.
k_{9a}	1 l/min	Release of VLDL from secretory pathway	Set to unity.
k_a	0.1497 l/min	Adipose FFA uptake of CM TAG (insulin independent)	Turnover of chylomicron TAG is estimated to have a half life of 4.5 min [21]. We use a 5 min half life giving a total degradation rate of $\log 2/5 = 0.1386$. The liver takes up gives 4% of this [23]. Bickerton [8] shows adipose tissue uptake 3 times more than skeletal muscle, therefore adipose uptake given by 72% of 0.1386.
k_{aa}	$3.11 \times 10^5\ \text{l}^2\ \text{mmol}^{-2}\ \text{min}^{-1}$	Adipose FFA esterification to TAG	Rate chosen such that the adipose FFA concentration is 0.57 mmol/l in the steady state.
k_{ai}	$2.08 \times 10^6\ \text{mmol}^{-1}$	Adipose FFA uptake of CM TAG (insulin dependent)	Negligible compared to insulin independent take up.
k_{al}	$0.00002\ \text{l}^2\ \text{mmol}^{-1}\ \text{min}^{-1}$	Pyruvate to ACoA conversion rate	Refs. [31], [32].
k_{ba}	0.0104 l/min	Adipose uptake of endogenous LP TAG	See description below
k_{bl}	0.156 l/min	Liver uptake of plasma NEFA	See description below
k_{bm}	0.226 l/min	Muscle uptake of plasma NEFA	See k_{bl}, muscle 50% of 0.4510.
k_{cl}	0.0075 l/min	Liver FFA Take-Up of chylomicron TAG	See k_a. Rate given by 4% of 0.1386.
k_{cm}	0.0449 l/min	Muscle FFA take up of chylomicron TAG	See k_a. Rate given by 24% of 0.1386.
k_d	1.733×10^8 l/min	Insulin degradation rate	The half life of insulin is approximately 4–6 min [15]. The degradation rate is calculated by (ln 2/half life). We use the lower bound of 4 min.
k_{dl}	3.5×10^8 l/mmol	Liver glycogenolysis (insulin inhibited rate)	See β_L.
k_{dy}	$4 \times 10^8\ \text{mmol}^{-1}$	Muscle glycogenolysis (insulin inhibited rate)	See β_M.
k_{ft}	$1.67 \times 10^{14}\ \text{mmol}^{-1}$	Adipose release of TAG to NEFA (insulin inhibited rate)	Approximated from data in Bickerton [8,9].
k_{ga}	$1.67 \times 10^6\ \text{mmol}^{-1}$	Adipose uptake of glucose	Rate chosen such that the adipose glucose steady state concentration is 2.54 mmol/l [56].
k_{gi}	$2.632 \times 10^7\ \text{mmol}^{-1}$	Insulin-stimulated transport of glucose from plasma to muscle.	Approximated using data from Kelley and Kraegen simultaneously [30,35].
k_{gl}	0.9277 l/min	Liver to blood glucose scalar	In combination with k_{gl2} this confers steady state concentrations of liver (8 mmol/l) to blood (5 mmol/l).
k_{gl2}	0.396 l/min	Blood to liver glucose scalar	In combination with k_{gl} this confers steady state concentrations of liver (8 mmol/l) to blood (5 mmol/l).
k_{gm}	0.0380 l/min	Blood to muscle glucose scalar	In combination with k_{gm2} this confers steady state concentrations of muscle (0.5 mmol/l) to blood (5 mmol/l).
k_{gm2}	0.0380 l/min	Muscle to blood glucose scalar	In combination with k_{gm} this confers steady state concentrations of muscle (0.5 mmol/l) to blood (5 mmol/l).
k_{gp}	0.311 l/min	Adipose glycerol return rate (to liver)	Rate chosen such that the glycerol steady state concentration is 0.17 mmol/l [31].
k_{lp}	0.25	Proportion of TAG released into the plasma from the action of LPL.	Estimated.
k_{LG}	8.95 mmol/l	Glucokinase affinity coefficient	Ref. [53]. Note, mouse data is used here.
k_{LH}	0.0115 mmol/l	Hexokinase affinity coefficient	Ref. [53]. Note, mouse data is used here.
k_{MH}	8.95 mmol/l	Hexokinase affinity coefficient	Same as k_{LG}.
k_{na}	0.0697	Adipose uptake of plasma NEFA	See k_{bl}, adipose 15.45% of (of 0.4510)
k_p	$1.41 \times 10^7\ \text{l}^2\ \text{mmol}^{-1}\ \text{min}^{-1}$	Pyruvate to ACoA conversion rate	Refs. [31],[32].

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Table 3 (*continued*)

Parameter	Value	Description	Source
k_{p6}	1.93×10^8 mmol^{-1}	Liver pyruvate to G-6-P (insulin inhibited)	Set such that gluconeogenesis and glycogenolysis produce equal quantities of glucose [59].
k_{pp}	0.5 l/min	Muscle pyruvate return to liver pyruvate	Rate chosen such that the muscle pyruvate concentration is at the given steady state.
k_r	0.00058 l/min	Liver uptake of endogenous LP TAG	See k_{ba}. Rate given by 4% of 0.0262.
k_{rep}	2.98 l/mmol	Hexokinase inhibition by G-6-P	Simple least squares fitting of experiments in Crane [10].
k_t	0.00348 l/min	Muscle uptake of endogenous LP TAG	See k_{ba}. Rate given by 24% of 0.0262.
k_{yl}	1.28×10^6 l^2 mmol^{-1} min^{-1}	Liver glycogenesis rate	
k_{ym}	21.364 l^2 mmol^{-1} min^{-1}	Muscle glycogenesis rate	Average of estimates in Price [48] and Shulman [51]. Rate is adjusted for glucose and insulin concentrations.
l_{max}	400 mmol/l	Maximum liver glycogen concentration	Ref. [55].
m_{max}	100 mmol/l	Maximum muscle glycogen concentration	Ref. [19].
v	7 mmol/l	Required glucose concentration for half maximal secretion of insulin	Obtained from a simple least squares data fit of experiments in Ref. [22] (Fig. 1A).
v_{10}	0.1 l/min	Max rate of hydrolysis of TAG to secretory pool	Set to release the same quantity of TAG as secreted in nascent VLDL particles.
v_{12}	40 mmol/l	Half maximal concentration for insulin's effect on switching the secretion of lipoprotein fractions	See k_{12}.
v_6	0.0158 l/min	Max rate of VLDL2 TAG secretion through secretory pathway	See v_9 description.
v_8	0.333 l/min	Max rate of FFA esterification to TAG	Comparatively very small, set to zero.
v_9	0.0159 l/min	Maximum rate for the bulk lipidation of VLDL	See additional parameter descriptions below.
v_{LG}	14.3 l/min	Glucokinase max rate	Estimated rate from Sreenan [53], scaled for 1.6 l liver volume. Note, mouse data is used here.
v_{LH}	5.57 l/min	Hexokinase max rate	Estimated rate from Sreenan [53], scaled for 1.6l liver volume. Note, mouse data is used here.
v_{MH}	54.288 l/min	Hexokinase max rate	See v_{LH}. Scaled for 16.4 l muscle volume.
y_0	0.1 mmol/l	Small parameters	Chosen to be $<O(0.01)$ of glycogen concentration.